

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 13-03-2026

Total Questions: 10 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. What is the first core design problem listed for Lexi? [2 Marks]
 - a) Formatting
 - b) Document Structur
 - c) Embellishing th
 - d) UI
2. AnswerRemember List three core design problems for Lexi as presented in the text. [5 Marks]
3. How does Lexi ensure its application code is not tied to a particular platform when supporting multiple look-and-feel standards? [2 Marks]
 - a) By hard-coding platform-specifi
 - b) widgets throughout th
 - c) application.
 - d) By making all UI widgets subclasses of
4. AnswerUnderstand Describe how the Bridge Pattern is applied in Lexi to support multiple window systems, mentioning the two main class hierarchies involved. [5 Marks]
5. AnswerUnderstand Explain the Decorator Pattern's approach to adding visual embellishments like borders and scroll bars to Lexi's UI dynamically, without creating a combinatorial explosion of subclasses. [5 Marks]

SECTION B: SHORT ANSWERS

6. Which design pattern is used to internally represent Lexi's document's physical structure? [2 Marks]
7. Which design pattern allows different formatting algorithms to be easily swapped in Lexi without changing the document structure? [2 Marks]
8. AnswerUnderstand Explain the main problem related to "Formatting" in Lexi's design and what solution is applied using a specific design pattern. [5 Marks]

9. AnswerUnderstand How does the Command Pattern enable multi-level undo/redo functionality in Lexi's user operations? [5 Marks]
10. AnswerUnderstand Describe the combined roles of the Iterator and Visitor patterns in Lexi for analytical operations like spelling checking and hyphenation, specifically addressing how they avoid changing Glyph classes. [5 Marks]
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